Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/05/2025
PI Name:	Blaber, Elizabeth Ph.D.		
Project Title:	Unraveling the Role of Mitochondrial Dysfunction and Senescence on Inhibition of Tissue Regeneration During Spaceflight and Amelioration by a Novel Countermeasure, PQQ		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	blabee@rpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	518-276-6548
Organization Name:	Rensselaer Polytechnic Institute		
PI Address 1:	Department of Biomedical Engineering		
PI Address 2:	JEC 7049 7th Fl		
PI Web Page:			
City:	Troy	**State:**	NY
Zip Code:	12180	**Congressional District:**	20
Comments:	NOTE: Moved to Rensselaer Polytechnic Institute in fall 2019 from NASA Ames Research Center.		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	05/01/2023	**End Date:**	04/30/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bai, Yang Ph.D. (Rensselaer Polytechnic Institute) Jonscher, Karen Ph.D. (University of Oklahoma, Oklahoma City) Singh, Keshav Ph.D. (University of Alabama, Birmingham)		
Grant/Contract No.:	80NSSC23K0775		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight factors, including microgravity and radiation, significantly affect tissue regeneration and repair in multiple physiological systems, including the skin, musculoskeletal, nervous, digestive, and immune systems. Our research, and that of others, has demonstrated activation of oxidative stress, inflammation, mitochondrial dysfunction, and senescence as key factors that contribute to the decline in tissue regenerative capacity in spaceflight, leading to tissue degeneration. In this proposal, we aim to investigate the effects of mitochondrial dysfunction and senescence on the ability of two key stem cell populations, skin and bone marrow stem cells, to conduct regenerative repair during spaceflight exposure. We specifically aim to interrogate the interplay between these two cellular mechanisms in the disruption of stem cell homeostasis, driving stem cell dysfunction, and inhibiting regeneration. Our research has also demonstrated that pyrroloquinoline quinone (PQQ), a potent dietary antioxidant, has anti-inflammatory and anti-senescent effects in multiple tissues. As PQQ also stimulates mitochondrial biogenesis and may be beneficial in diseases associated with mitochondrial dysfunction, including impaired tissue regeneration, we aim to test the ability of this countermeasure to ameliorate spaceflight effects on stem cell-based tissue regeneration. We specifically hypothesize that spaceflight factors cause systematic mitochondrial dysfunction and inflammatory responses, resulting in cellular senescence and tissue degeneration in mammalian tissues that can be mitigated by PQQ. We will use unique animal models, including a mtDNA-depleter model developed by Dr. Singh; and a senolytic mouse model (CDKN1a/p21-/--p16-3mR) developed by Dr. Blaber, in collaboration with Dr. Judith Campisi; that will enable direct interrogation of the systemic regulators, including oxidative stress and inflammation, mediating mitochondrial dysfunction, and senescence in stem cell populations. Furthermore, we will model regenerative repair through a full dermal thickness excisional wound, induced during exposure to simulated spaceflight exposure (hindlimb unloading and 0.5 Gy simplified Galactic Cosmic Ray simulation (simGCRsim)), enabling study of regenerative mechanisms under homeostatic conditions as well as during functional repair following injury. These studies, combined with an array of advanced analysis techniques, including spatial transcriptomics, single cell RNA sequencing, Luminex based cytokine and inflammation assays, and ex-vivo functional stem cell assays, will enable us to determine the cellular and molecular mechanisms underpinning the deleterious impact of spaceflight induced mitochondrial dysfunction on stem-cell based tissue regeneration in two uniquely regenerating tissues. Finally, by testing efficacy of a dietary countermeasure (approved for use in humans) to promote wound healing under spaceflight conditions, this study is aligned with NASA's Thrive In DEep Space (TIDES) research initiative. [Ed. Note: The goal of the NASA Space Biology TIDES initiative is to enable long-duration missions and improve life on Earth through innovative research: https:// .]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Mechanobiological and physiological consequences of long-duration spaceflight are of increasing concern for human space exploration. Our work, and that of others, has demonstrated that spaceflight factors significantly impair somatic stem cell-based tissue regeneration, leading to degenerative decline in highly regenerative tissues (e.g., skin and musculoskeletal systems), that is similar to degeneration that occurs during aging. Other factors affecting stem cell function during regeneration and repair may include oxidative stress, inflammation, DNA damage, mitochondrial dysfunction, and senescence induction, all of which are influenced by spaceflight. Recent reports suggest that, during spaceflight, humans and other species accumulate mitochondrial damage, resulting in mitochondrial dysfunction, which may exacerbate senescence in key stem cell populations. Pyrroloquinoline quinone (PQQ), a potent dietary antioxidant, has anti-inflammatory and anti-senescent effects in multiple tissues, stimulates mitochondrial biogenesis, and may be beneficial in diseases associated with mitochondrial dysfunction. This proposal will therefore study the systemic and mechanistic basis for spaceflight-induced tissue degeneration and regenerative repair, and will test a novel dietary countermeasure to ameliorate these effects. We aim to comprehensively elucidate cellular and molecular mechanisms underpinning the deleterious impact of spaceflight-induced mitochondrial dysfunction and senescence on stem cell-based tissue regeneration in two highly regenerative and inter-connected tissues – skin and bone. We will use innovative, inducible rodent models and a potent, dietary antioxidant, shown to be safe in humans, to identify a novel countermeasure that may preserve tissue health in long-duration space exploration. During the last performance period, we have specifically focused on breeding of mice and experimental planning for studies at the NASA Space Radiation Laboratory (NSRL). Additionally, NSRL proposals for the next two campaigns were submitted in order to conduct our studies. Finally, we optimized wound healing assays, both in vivo and in vitro, and began characterization of bone marrow fibroblast dynamics following inflammation/oxidative stress.
Bibliography Type:	Description: (Last Updated: 04/03/2025)